

# Bifurcations of Hamiltonian Periodic Orbits

---

Dongho Lee

December 12th, 2025

SNU, QSMS

Presented in Symplectic Learning Seminar, SNU



Download slides

## Well-known Classical Fact from 70s

There are 8 types of *generic* bifurcation of Hamiltonian periodic orbits.

## Well-known Classical Fact from 70s

There are 8 types of *generic* bifurcation of Hamiltonian periodic orbits.

## Contents

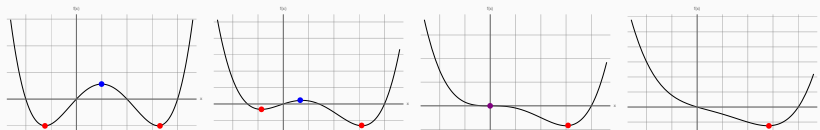
- Dynamical Systems and Bifurcations
- Poincaré Section and Stability
- Bifurcation and Symplectic Homology
- Ingredients : Classical works of Lyapunov and Meyer
- Classification of Generic Bifurcations

Main references : [Mey70], [AM78]

# **Dynamical Systems and Bifurcations**

---

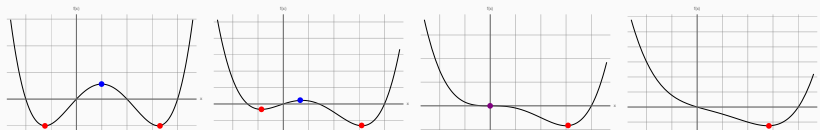
## Toy Example : Critical Points of a Function



**Figure 1:** Graphs of  $f_t(x) = x(x^2 - t)(x - 1)$  for varying  $t$ .

- $t < 0$ , two minima (●) and one maximum. (●)
- $t = 0$ , ● and ● meets at the inflection point. (●)
- $t > 0$ , ● vanishes and one minimum survives. (●)

## Toy Example : Critical Points of a Function

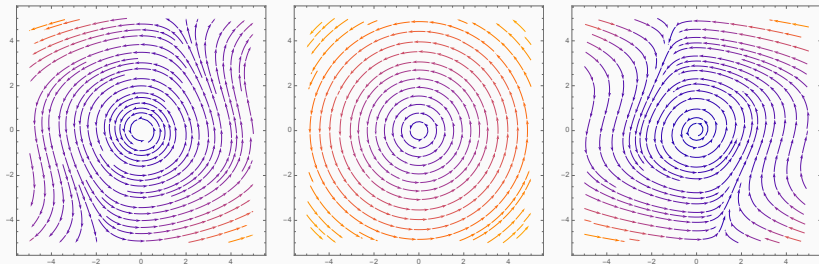


**Figure 1:** Graphs of  $f_t(x) = x(x^2 - t)(x - 1)$  for varying  $t$ .

- $t < 0$ , two minima (●) and one maximum. (●)
- $t = 0$ , ● and ● meets at the inflection point. (●)
- $t > 0$ , ● vanishes and one minimum survives. (●)

**Idea.** Similar thing happens for periodic orbits.

## Example : Van der Pol Oscillator



**Figure 2:** Phase portraits of Van der Pol oscillator for varying  $\mu$ .

Van der Pol oscillator is given by

$$\dot{x} = \mu(1 - y^2)x - y, \quad \dot{y} = x.$$

This is an example of a phenomenon called Hopf bifurcation.

# Dynamical System

According to [Sma67], a **dynamical system** consists of following data.

- A space  $W$  (we use smooth manifolds)
- A group action  $G \rightarrow \text{End}(W)$  (we use diffeomorphisms)



# Dynamical System

According to [Sma67], a **dynamical system** consists of following data.

- A space  $W$  (we use smooth manifolds)
- A group action  $G \rightarrow \text{End}(W)$  (we use diffeomorphisms)

If  $G = \mathbb{Z} = \langle F \rangle$ , we say the system is **discrete**.

According to [Sma67], a **dynamical system** consists of following data.

- A space  $W$  (we use smooth manifolds)
- A group action  $G \rightarrow \text{End}(W)$  (we use diffeomorphisms)

If  $G = \mathbb{Z} = \langle F \rangle$ , we say the system is **discrete**.

## Interesting objects

- **Fixed point**  $x \in W$  such that  $F(x) = x$ .
- **Periodic point**  $x \in W$  such that  $F^N(x) = x$  for some  $N$ .
- **Invariant subspace**  $S \subset S$  such that  $F(S) \subset S$
- Non-wandering points, recurrent points, etc.

# Continuous Dynamical System

If  $G = \mathbb{R} = \{F_t\}$ , we say the system is **continuous**.

A vector field  $\frac{d}{dt}F_t|_{t=0} = X$  consists equivalent information with  $F_t$ , which is called the infinitesimal generator of  $F_t$ .

**Example.** The system determined by an ODE  $\dot{x} = X(x)$ .

# Continuous Dynamical System

If  $G = \mathbb{R} = \{F_t\}$ , we say the system is **continuous**.

A vector field  $\frac{d}{dt}F_t|_{t=0} = X$  consists equivalent information with  $F_t$ , which is called the infinitesimal generator of  $F_t$ .

**Example.** The system determined by an ODE  $\dot{x} = X(x)$ .

## Interesting objects

- **Critical points**  $x \in W$  such that  $F_t(x) = x$  for any  $t$ . (or  $X_x = 0$ .)
- **Periodic orbits**  $x \in W$  such that  $F_\tau(x) = x$  for some  $\tau > 0$ .
- **Invariant subsets**  $S \subset W$  such that  $F_t(S) \subset S$  for any  $t$  (or  $t > 0$ ).

# Hamiltonian Dynamical System

Let  $(W, \omega)$  be a symplectic manifold,  $H : W \rightarrow \mathbb{R}$ .

$H$  defines a **Hamiltonian vector field**  $X_H$  via relation

$$i_{X_H}\omega = \omega(X_H, -) = -dH.$$

We call  $(W, \omega, H)$  a **Hamiltonian dynamical system**.

- $X_H$  preserves  $\omega$ , i.e.  $\mathcal{L}_{X_H}\omega = 0$ .
- $X_H$  preserves  $H$ , i.e.  $X_H$  flows on  $H^{-1}(c)$  for each  $c$ .

# Hamiltonian Dynamical System

Let  $(W, \omega)$  be a symplectic manifold,  $H : W \rightarrow \mathbb{R}$ .

$H$  defines a **Hamiltonian vector field**  $X_H$  via relation

$$i_{X_H}\omega = \omega(X_H, -) = -dH.$$

We call  $(W, \omega, H)$  a **Hamiltonian dynamical system**.

- $X_H$  preserves  $\omega$ , i.e.  $\mathcal{L}_{X_H}\omega = 0$ .
- $X_H$  preserves  $H$ , i.e.  $X_H$  flows on  $H^{-1}(c)$  for each  $c$ .

The first example is the geodesic flow on a Riemannian manifold  $(M, g)$ ,

$$H_g(q, p) = \frac{|p|_g^2}{2}$$

which is defined on  $T^*M$ .

## Example : Classical Mechanics

Consider the mechanical system in  $T^*\mathbb{R}^n$  with energy

$$H(q, p) = \frac{|p|^2}{2} + V(q).$$

Corresponding Hamiltonian vector field is

$$X_H = p \frac{\partial}{\partial q} - \nabla V \frac{\partial}{\partial p},$$

which can be translated into

$$\dot{q} = p, \quad \dot{p} = \ddot{q} = -\nabla V.$$

This is Newton's 2nd law, which governs the classical mechanics.

# Structural Stability

Two vector fields  $X, Y$  on  $W$  are **topologically conjugate** if there exists a homeomorphism  $F : W \rightarrow W$  which carries the oriented orbits of  $X$  to the ones of  $Y$ .



# Structural Stability

Two vector fields  $X, Y$  on  $W$  are **topologically conjugate** if there exists a homeomorphism  $F : W \rightarrow W$  which carries the oriented orbits of  $X$  to the ones of  $Y$ .

If  $X$  has a neighborhood  $\mathcal{U} \subset \mathcal{X}(W)$  such that if  $Y \in \mathcal{U}$  then  $X$  and  $Y$  are topologically conjugate, we call  $X$  is **structurally stable**.

# Structural Stability

Two vector fields  $X, Y$  on  $W$  are **topologically conjugate** if there exists a homeomorphism  $F : W \rightarrow W$  which carries the oriented orbits of  $X$  to the ones of  $Y$ .

If  $X$  has a neighborhood  $\mathcal{U} \subset \mathcal{X}(W)$  such that if  $Y \in \mathcal{U}$  then  $X$  and  $Y$  are topologically conjugate, we call  $X$  is **structurally stable**.

**Idea.** A small perturbation of topologically stable vector field  $X$  cannot change the dynamical system essentially.

# Bifurcation

Consider a family of vector fields  $X_t$  on  $W$ , parametrized by a set  $0 \in I$ .

A point  $0 \in I$  is called

- a **robust point** if there exists a neighborhood  $U$  of  $0$  in  $I$  such that if  $c \in U$ ,  $X_0$  and  $X_c$  are topologically conjugate.
- a **bifurcation point** if not.

We will only consider 1-dimensional bifurcation, i.e.  $I$  is 1-manifold.

# Bifurcation

Consider a family of vector fields  $X_t$  on  $W$ , parametrized by a set  $0 \in I$ .

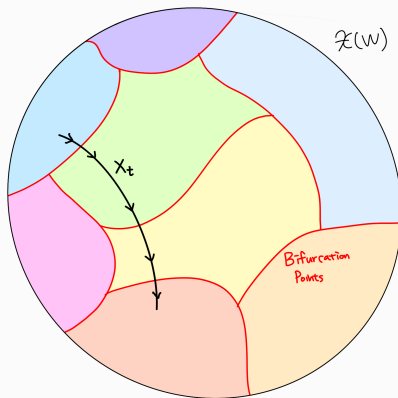
A point  $0 \in I$  is called

- a **robust point** if there exists a neighborhood  $U$  of  $0$  in  $I$  such that if  $c \in U$ ,  $X_0$  and  $X_c$  are topologically conjugate.
- a **bifurcation point** if not.

We will only consider 1-dimensional bifurcation, i.e.  $I$  is 1-manifold.

**Idea.** If we can find *generic* properties of vector fields, we can talk about *generic* bifurcations.

# Bifurcation



**Figure 3:** Very schematic picture of a bifurcation.

# Poincaré Section and Stability

---

## Theorem (Poincaré Section)

*Let  $\gamma$  be a periodic orbit of  $X$  in  $W$  and  $x \in \gamma$ .*

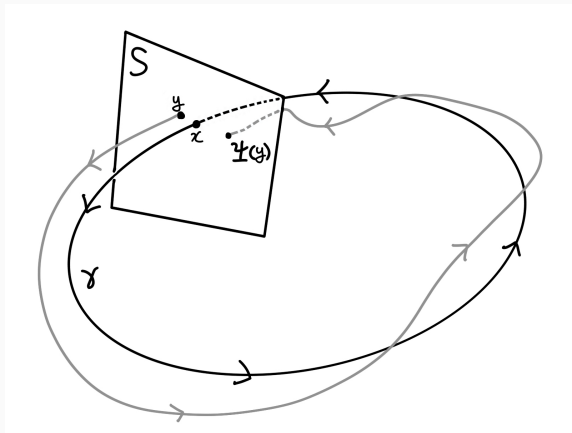
- 1. There exists a codimension 1 submanifold  $S \subset W$  such that  $x \in S$  and is transversal to  $X$ .*
- 2. Define a map  $\Psi : S \rightarrow W$  by*

$$\Psi(y) = Fl_{\tau(y)}^X(y)$$

*where  $\tau(y) = \min\{t > 0 : Fl_t^X(y) \in S\}$ , then this is well-defined and unique up to conjugation.*

We call  $S$  **Poincaré section**, and  $\Psi$  **return map** or **Poincaré map**.

# Poincaré Section



**Figure 4:** A Poincaré section and the return map



# Hamiltonian Version of Poincaré Section

## Proposition

*Let  $X = X_H$  be a Hamiltonian vector field and  $\gamma$  be a periodic orbit with regular energy  $c_0$ . There exists a Poincaré section  $Y$  such that*

- 1.  $Y \subset W$  is a contact submanifold.*
- 2.  $S_c = Y \cap H^{-1}(c)$  for  $c$  close to  $c_0$  is a symplectic submanifold.*
- 3. The return map on  $S_c$  is a symplectomorphism.*

# Hamiltonian Version of Poincaré Section

## Proposition

*Let  $X = X_H$  be a Hamiltonian vector field and  $\gamma$  be a periodic orbit with regular energy  $c_0$ . There exists a Poincaré section  $Y$  such that*

- 1.  $Y \subset W$  is a contact submanifold.*
- 2.  $S_c = Y \cap H^{-1}(c)$  for  $c$  close to  $c_0$  is a symplectic submanifold.*
- 3. The return map on  $S_c$  is a symplectomorphism.*

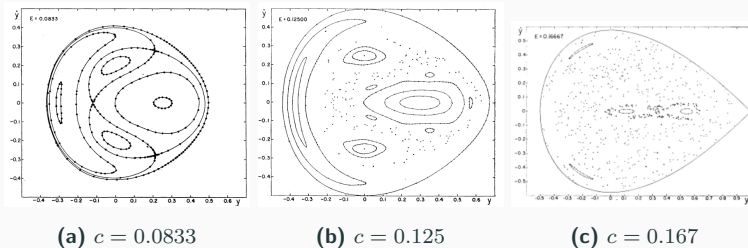
$2n$ -dimensional  $W$   $\Leftrightarrow$   $(2n - 2)$ -dimensional  $S$

Flow of vector field  $X_H$   $\Leftrightarrow$  Return map  $\Psi$

1-periodic orbit of  $X_H$  on  $W$   $\Leftrightarrow$  a fixed point of  $\Psi$  on  $S$

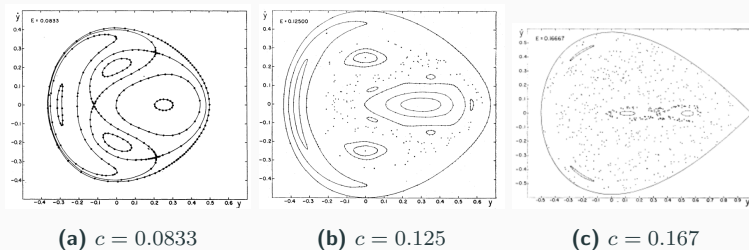
$k$ -periodic orbit of  $X_H$   $\Leftrightarrow$   $k$ -family of  $k$ -periodic points of  $\Psi$ .

## Aside : A Standard Picture the Poincaré Section



**Figure 5:** [HH64] Poincaré sections of the Hénon-Heiles system

## Aside : A Standard Picture the Poincaré Section



**Figure 5:** [HH64] Poincaré sections of the Hénon-Heiles system

1. Take an initial condition  $x$  and make orbit  $\gamma$ .
2. Mark the intersections  $x_i$ 's of  $\gamma$  and  $S$ .
3. Draw a curve (if possible) between  $x_i$ 's.
4. Repeat for different initial conditions.

# Characteristic Exponents of Critical Points

Let  $x$  be a critical point of  $X_H$  on  $W$ .

The **characteristic exponents** of  $x$  are the eigenvalues of  $dX_H(p)$ .

In symplectic case, the exponents come with pairs,  $\pm\alpha$  and  $\pm\bar{\alpha}$ .

# Characteristic Exponents of Critical Points

Let  $x$  be a critical point of  $X_H$  on  $W$ .

The **characteristic exponents** of  $x$  are the eigenvalues of  $dX_H(p)$ .

In symplectic case, the exponents come with pairs,  $\pm\alpha$  and  $\pm\bar{\alpha}$ .

Let  $x$  be a critical point, and  $\{\alpha_i\}$  be characteristic exponents.

Let's denote  $\lambda_i = \exp(\alpha_i)$  for the future consistency.

- If  $\lambda_i \notin S^1$  for any  $i$ , call  $x$  **hyperbolic**.
- If  $\lambda_i \in S^1$  for any  $i$ , call  $x$  **elliptic**.

# Elliptic and Hyperbolic Critical Points

Let  $\alpha$  be a characteristic exponent of  $x$ ,  $\lambda = e^\alpha$  and  $v$  be eigenvector.

Consider the linearized flow in direction  $v$  at a critical points.

- If  $|\lambda| > 1$ , the flow is expanding and tends away from  $x$ . (**unstable**)
- If  $|\lambda| < 1$ , the flow is contracting and tends closer to  $x$ . (**stable**)
- If  $|\lambda| = 1$ , the flow rotates around the  $x$ . (**center**)

In Hamiltonian case, the stable and unstable dimensions are the same.

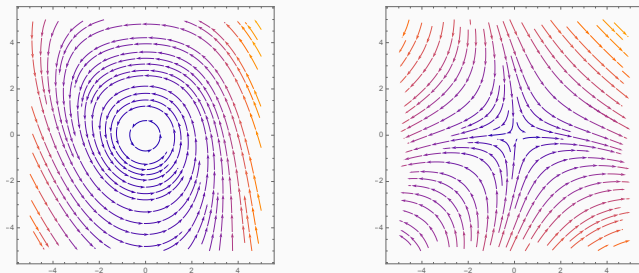
# Elliptic and Hyperbolic Critical Points

Let  $\alpha$  be a characteristic exponent of  $x$ ,  $\lambda = e^\alpha$  and  $v$  be eigenvector.

Consider the linearized flow in direction  $v$  at a critical points.

- If  $|\lambda| > 1$ , the flow is expanding and tends away from  $x$ . (**unstable**)
- If  $|\lambda| < 1$ , the flow is contracting and tends closer to  $x$ . (**stable**)
- If  $|\lambda| = 1$ , the flow rotates around the  $x$ . (**center**)

In Hamiltonian case, the stable and unstable dimensions are the same.



**Figure 6:** Elliptic and hyperbolic fixed points



# Characteristic Multipliers of Periodic Orbits

Let  $\gamma$  be a periodic orbit,  $S$  be a Poincaré section and  $\gamma \cap S = \{x\}$ .

The **characteristic multipliers** of  $\gamma$  are the eigenvalues of  $d\Psi_x$ .

The multipliers come with pairs,  $\lambda^{\pm 1}$ ,  $\bar{\lambda}^{\pm 1}$ .

# Characteristic Multipliers of Periodic Orbits

Let  $\gamma$  be a periodic orbit,  $S$  be a Poincaré section and  $\gamma \cap S = \{x\}$ .

The **characteristic multipliers** of  $\gamma$  are the eigenvalues of  $d\Psi_x$ .

The multipliers come with pairs,  $\lambda^{\pm 1}$ ,  $\bar{\lambda}^{\pm 1}$ .

If  $\dim S = 2$ , which is of our interest, it simplifies by  $\lambda \in S^1 \cup \mathbb{R}_{|\cdot|>1}$ .

We choose **principal multiplier**  $\lambda_\gamma$  among  $\pm 1$ ,  $\text{Im}\lambda_\gamma > 0$  and  $|\lambda_\gamma| > 1$ .

- If  $\lambda_\gamma \in \mathbb{R}_{|\cdot|>1}$ , call  $\gamma$  **hyperbolic**.
- If  $\lambda_\gamma \in S^1 \setminus \{\pm 1\}$ , call  $\gamma$  **elliptic**.

## Proposition ([BL34] Birkhoff-Lewis)

*Let  $\Psi : U \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be a symplectomorphism with  $\Psi(0) = 0$ .*

*If  $\Psi$  is an elementary twist mapping (See Appendix 1), then in every neighborhood of 0,*

- 1. there exists an invariant circle in  $U$  without periodic points.*
- 2. for any integer  $k$ , there exists a  $k$ -periodic point in  $U$ .*

It means the elliptic periodic orbits can be detected from nearby orbits, while the hyperbolic ones cannot be.

In this sense, the hyperbolic periodic orbits are *qualitatively invisible*.

## Aside : Vague Attractor of Kolmogorov

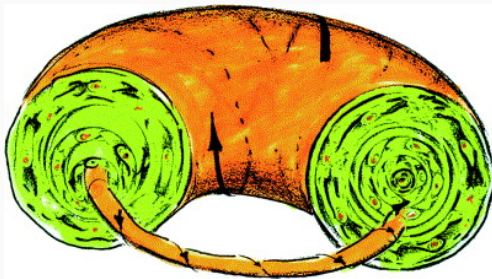


Figure 7: [EI 03] VAK

We say such configuration a **vague attractor of Kolmogorov**, or **VAK**.

# **Bifurcation and Symplectic Homology**

---

# Conley-Zehnder Indices

Let  $\gamma$  be a periodic Hamiltonian orbit of period  $\tau$ ,  $S$  be a Poincaré section,  $\gamma \cap S = \{x\}$  and  $\Psi$  be the return map.

$\gamma$  is **non-degenerate** if  $\det(d\Psi|_x - \text{Id}) \neq 0$ , i.e. the linearized return map doesn't have 1 as an eigenvalue.

# Conley-Zehnder Indices

Let  $\gamma$  be a periodic Hamiltonian orbit of period  $\tau$ ,  $S$  be a Poincaré section,  $\gamma \cap S = \{x\}$  and  $\Psi$  be the return map.

$\gamma$  is **non-degenerate** if  $\det(d\Psi|_x - \text{Id}) \neq 0$ , i.e. the linearized return map doesn't have 1 as an eigenvalue.

The **Conley-Zehnder index** is an integer assigned to  $\gamma$ , defined by

$$\gamma_{CZ}(\gamma) = \frac{1}{2}\text{Sign}Q_0 + \sum \text{Sign}Q_t + \frac{1}{2}\text{Sign}Q_\tau \in \mathbb{Z}$$

where  $Q_t$  is a quadratic form  $Q_t(v, v) = \omega(v, \dot{\Psi}(t)v)$  on  $\ker(\Psi(t) - \text{Id})$ , defined for  $t$  such that  $\det(\Psi(t) - \text{Id}) = 0$ , and  $\text{Sign}$  is the signature of a quadratic form.

# (Very Brief) Symplectic Homology

$H : W \rightarrow \mathbb{R}$  be a Hamiltonian,  $c$  be a regular value.

Assume :  $H^{-1}(c)$  is a contact type hypersurface, consists only of non-degenerate periodic orbits.

$\Rightarrow$  We can define the **symplectic homology**  $SH^{S^1,+}(W_c)$ ,

- generated by periodic Hamiltonian orbits on  $H^{-1}(c)$ ,
- graded by Conley-Zehnder indices,
- Differentials are given by Floer cylinders.



# Relation between Symplectic Homology and Bifurcation

If  $\text{CritVal}(H) \cap [c_1, c_2] = \emptyset$ , then  $SH^{S^1,+}(W_{c_1}) \simeq SH^{S^1,+}(W_{c_2})$ .

$\Rightarrow$  CZ-indices of *all* periodic orbits are topologically controlled.

# Relation between Symplectic Homology and Bifurcation

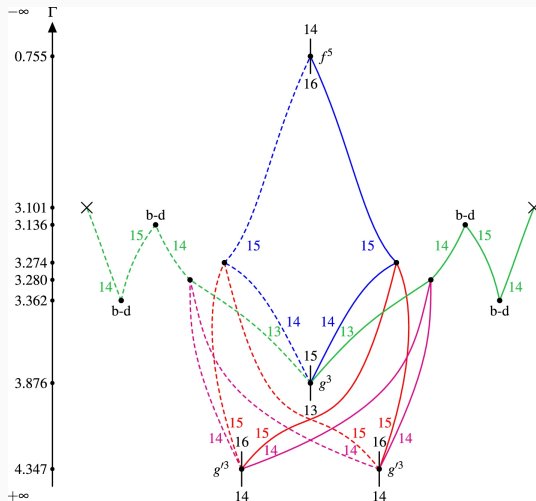
If  $\text{CritVal}(H) \cap [c_1, c_2] = \emptyset$ , then  $SH^{S^1,+}(W_{c_1}) \simeq SH^{S^1,+}(W_{c_2})$ .

$\Rightarrow$  CZ-indices of *all* periodic orbits are topologically controlled.

Assume that we're away from the critical energies.

- Assume that there exists a family of periodic orbits  $\gamma_c$  parametrized by the energy  $c$ . (an *orbit cylinder*)
- If  $\gamma_c$  is non-degenerate for  $c_1 \leq c \leq c_2$ , then  $\mu_{CZ}(\gamma_{c_1}) = \mu_{CZ}(\gamma_{c_2})$ .
- Once  $\gamma_c$  degenerates,  $\mu_{CZ}(\gamma_c)$  might change, or the family might just vanish away.
- Since  $SH^{S^1,+}(W_c)$  must be preserved, other orbits must take the charge of the changed index by being born, vanishing, or changing its own index.

# Example : Hill's Lunar Problem



**Figure 8:** [Ayd23] Bifurcation diagram of Hill's lunar problem

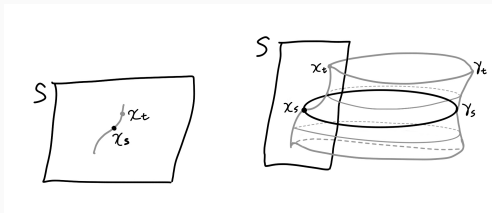
## **Ingredients : Classical Works**

---

# Orbit Cylinder

## Lemma (Orbit Cylinder)

If  $\lambda_x \neq 1$ , there exists neighborhoods  $x_s \in U \subset M$  and  $x \in V \subset S^1$  and a smooth map  $P : V \rightarrow U$  such that  $P(s) = x$  and  $\{(P(t), t) : t \in V\}$  consists of fixed points.



In dimension 4, this can be interpreted as a family of orbits, and the *height* parameter  $t$  can be chosen as the energy.

**Note.** This theorem holds for arbitrary dimension.

# Lyapunov's Theorem

Let  $x$  be a critical point of  $X_H$ .  $x$  is called  $\mathcal{H}$ -**elementary** if

1. 0 is not a characteristic exponent.
2. If  $\alpha \in i\mathbb{R}$  is a characteristic exponent,  $\alpha$  has multiplicity 1.
3. If  $\alpha, \beta \in i\mathbb{R}_{>0}$  are characteristic exponents, they are  $\mathbb{Z}$ -linearly independent.

# Lyapunov's Theorem

Let  $x$  be a critical point of  $X_H$ .  $x$  is called  $\mathcal{H}$ -**elementary** if

1. 0 is not a characteristic exponent.
2. If  $\alpha \in i\mathbb{R}$  is a characteristic exponent,  $\alpha$  has multiplicity 1.
3. If  $\alpha, \beta \in i\mathbb{R}_{>0}$  are characteristic exponents, they are  $\mathbb{Z}$ -linearly independent.

## Theorem ([Buc70])

*For generic Hamiltonian  $H$ , every critical point of  $X_H$  is  $\mathcal{H}$ -elementary.*

# Lyapunov's Theorem

## Theorem (Lyapunov)

*Let  $x$  be an  $\mathcal{H}$ -elementary critical point of  $X_H$ . If  $i\beta$  is a characteristic exponent for  $\beta \in \mathbb{R}$ , there exists a 2-dimensional submanifold  $C_\beta$  such that*

- 1.  $x \in C_\beta$ .*
- 2.  $C_\beta$  is  $X_H$ -invariant submanifold.*
- 3.  $T_x C_\beta$  is an eigenspace corresponds to  $\pm i\beta$ .*
- 4.  $C_\beta$  is a union of periodic orbits  $\gamma_r$  which converges to  $x$  as  $r \rightarrow 0$ .*
- 5. The period of  $\gamma_r$  converges to  $2\pi/\beta$  as  $r \rightarrow 0$ .*



# Lyapunov's Theorem

## Theorem (Lyapunov)

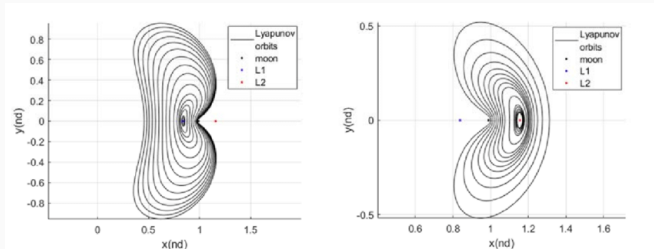
*Let  $x$  be an  $\mathcal{H}$ -elementary critical point of  $X_H$ . If  $i\beta$  is a characteristic exponent for  $\beta \in \mathbb{R}$ , there exists a 2-dimensional submanifold  $C_\beta$  such that*

- 1.  $x \in C_\beta$ .*
- 2.  $C_\beta$  is  $X_H$ -invariant submanifold.*
- 3.  $T_x C_\beta$  is an eigenspace corresponds to  $\pm i\beta$ .*
- 4.  $C_\beta$  is a union of periodic orbits  $\gamma_r$  which converges to  $x$  as  $r \rightarrow 0$ .*
- 5. The period of  $\gamma_r$  converges to  $2\pi/\beta$  as  $r \rightarrow 0$ .*

Simply, there exist families of periodic orbits emerging from  $x$  for each elliptic exponent.

# Lyapunov Orbits

**Example.** Lyapunov orbits of the restricted three-body problem, emerges from the Lagrange points.



**Figure 9:** [ZWZZ20] Lyapunov orbits of RTBP, associated to  $L_1$  and  $L_2$ .

# Generating Function

Consider  $X_H$  on  $W^4$ , and go to the 2-dimensional section  $M$ .

Let  $\varphi : M \times S$  be a 1-parameter family of symplectomorphisms, and  $x$  be a fixed point of  $\varphi_s$ . We simply say  $(x, s)$  is a fixed point of  $\varphi_t$ .

Locally, we take a fixed point  $x = (0, 0)$  and denote  $\varphi_s$  as

$$Q = Q(q, p), \quad P = P(q, p).$$

[Poi87] observed that

$$\Omega = (P - p)d(Q + q) - (Q - q)d(P + p)$$

is closed, so there exists a function  $G$  such that  $dG = \Omega$ .

We call  $G$  a **generating function**.

# Generating Function

Assume that  $\lambda \neq -1$ , then under new coordinates

$$\xi = Q + q, \quad \eta = P + p,$$

we have

$$\frac{\partial G}{\partial \xi} = P - p, \quad \frac{\partial G}{\partial \eta} = q - Q,$$

and the fixed point of  $(Q, P)$  corresponds to the critical point of  $G$ .

# Generating Function

Assume that  $\lambda \neq -1$ , then under new coordinates

$$\xi = Q + q, \quad \eta = P + p,$$

we have

$$\frac{\partial G}{\partial \xi} = P - p, \quad \frac{\partial G}{\partial \eta} = q - Q,$$

and the fixed point of  $(Q, P)$  corresponds to the critical point of  $G$ .

- Hyperbolic fixed point of  $\varphi_s$  corresponds to a saddle point of  $G$ .
- Elliptic fixed point of  $\varphi_s$  corresponds to maximum or minimum of  $G$ .

(See Appendix 2 for the reason.)

# Meyer's Theorem - Extremal Fixed Point

We call  $(x, s)$  is **extremal fixed point** if, for generating function  $G = G(\xi, \eta, t)$ , at  $(0, 0, 0)$ ,

$$G = G_\xi = G_\eta = G_{\eta\eta} = G_{\eta\xi} = 0, \\ G_{\xi\xi} \neq 0, \quad G_{\eta t} \neq 0, \quad G_{\eta\eta\eta} \neq 0.$$

**Note.** This is the case  $\lambda_x = 1$ .

## Theorem ([Mey70])

*Let  $(x, s)$  be an extremal fixed point. Then,*

- 1. There exists a family of fixed points  $x_t$ .*
- 2.  $x$  divides  $x_t$  into two arcs: one entirely consists of hyperbolic fixed points and the other consists of elliptic fixed points.*
- 3. In any coordinate system,  $s$  is a nondegenerate maximum or minimum of the family.*

# Outline of Proof

We have an equation  $G_\eta = G_\xi = 0$  at  $(0, 0, 0)$ , and the Jacobian is

$$\begin{pmatrix} 0 & 0 & G_{\eta t} \\ 0 & G_{\xi\xi} & G_{t\xi} \end{pmatrix}.$$

This has rank 2, so we have solution  $(\xi, \eta, t) = (\xi(s), s, t(s))$  for the equation, i.e.

$$G_\xi(\xi(s), s, t(s)) = G_\eta(\xi(s), s, t(s)) = 0.$$

From this, we can deduce that  $t'(0) = 0$  and  $t''(0) \neq 0$ , and

$$\begin{pmatrix} G_{\xi\xi} & G_{\eta\xi} \\ G_{\xi\eta} & G_{\eta\eta} \end{pmatrix} (\xi(s), s, t(s)) = \begin{pmatrix} G_{\xi\xi}(0) & 0 \\ 0 & sG_{\eta\eta\eta}(0) \end{pmatrix} + (\text{h.o.t})$$

# Meyer's Theorem - Transitional Fixed Point

We call  $(x, s)$  is a **transitional fixed point** if  $\lambda_x = -1$  and satisfies some non-degeneracy condition. (See Appendix 3)

## Theorem ([Mey70])

*Let  $(x, s)$  be a transitional fixed point and  $x_t$  be an orbit cylinder in  $U \times V \subset M \times S^1$ . Then we can choose  $V$  small enough so that  $V \setminus \{s\} = V_H \cup V_E$ , where  $x_s$  is hyperbolic (elliptic) if  $s \in V_H$  ( $V_E$ ). Moreover, either one of the following holds.*

- 1. There exists functions  $x_1^H, x_2^H : V_H \rightarrow U$  such that  $(x_i^H(t), t)$  is elliptic periodic points of period 2 and  $x_i^H(t) \rightarrow x$  as  $t \rightarrow s$ ,*
- 2. There exists functions  $x_1^E, x_2^E : V_E \rightarrow U$  such that  $(x_i^E(t), t)$  is hyperbolic periodic points of period 2 and  $x_i(t) \rightarrow x$  as  $t \rightarrow s$ .*

We call these **an arc of nearby hyperbolic/elliptic periodic points**.



# Meyer's Theorem - Bifurcation Points

Let  $CM(x) = \{\exp(\pm 2\pi i l/k)\}$ ,  $(l, k) = 1$  and  $0 < l < k$ .

We call  $(x, s)$  is a  **$k$ -bifurcation point** if it satisfies some non-degeneracy condition. (See Appendix 3.)

## Theorem ([Mey70])

*Let  $(x, s)$  be  $k$ -bifurcation point. Then for small orbit cylinder  $U \times V$ ,*

- 1. ( $k = 3$ ) There exists three arcs of nearby hyperbolic periodic points of period 3, parametrized by  $V \setminus \{s\}$ , converging to  $x$ .*
- 2. ( $k \geq 5$ ) There exists  $k$  arcs of nearby hyperbolic periodic points and  $k$  arcs of nearby elliptic periodic points of period  $k$ , parametrized by  $V_+ \subset V \setminus \{s\}$ , converging to  $x$ .*
- 3. ( $k = 4$ ) One of ( $k = 3$ ) or ( $k \geq 5$ )-phenomena occurs.*

Let  $\mathcal{G}$  be the set of 1-parameter family of symplectomorphisms on  $M$  such that

1. Each periodic point is elliptic, hyperbolic, extremal or transitional.
2. If  $CM(x) = \{\exp(\pm 2\pi i l/k)\}$  for  $k \geq 3 \leq N$ , then  $(x, s)$  is  $k$ -bifurcation point.

## Theorem ([Mey70])

*$\mathcal{G}$  is a residual set in the space of 1-parameter family of symplectomorphisms.*

$\Rightarrow$  Now we know the *generic* bifurcation behavior.

# **Classification of Generic Bifurcations**

---

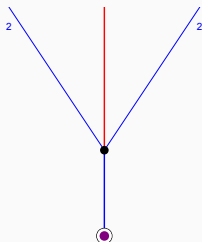
# How to classify bifurcation

Let  $\dim W = 4$ , so the Poincaré section is 2-dimensional.

- For critical points, characteristic exponents are  $\{\pm\alpha, \pm\bar{\alpha}\}$  in general.
- When there is  $\alpha \in i\mathbb{R}$ , Lyapunov theorem can be applied.
- For periodic orbits, if  $\lambda_\gamma \neq 1$ , there is an orbit cylinder.
- If  $\lambda_\gamma^k = 1$ ,  $k$ -th cover of the orbit bifurcates.
- Using Poincaré section, a set of  $k$  periodic points of period  $k$  corresponds to a periodic orbit of period  $k$ .
- The nearby hyperbolic/elliptic arc on the Poincaré section corresponds to the orbit cylinder of hyperbolic/elliptic orbits.

# Bifurcation Diagram

We'll see this kind of diagrams.



- Vertical axis is the energy.
- $\bullet$  is the bifurcation point.
- $\odot$  is the critical point of  $H$ .
- Each line is a family of periodic points of the return map in Poincaré section.
- Thick line is a family of fixed points, or 1-periodic orbits.
- Thin line is a family of  $k$ -periodic orbit, where the period is written aside the line. (2 here.)
- —,  $\odot$  are elliptic, —,  $\odot$  are hyperbolic, and  $\odot$  is saddle.

# List of Generic Bifurcations

| No.  | Name                      | Case                                             | Tool                                       |
|------|---------------------------|--------------------------------------------------|--------------------------------------------|
| I    | Burst / Reincarnation     | $x$ with<br>$i\alpha \in CE(x)$                  | Lyapunov theorem                           |
| II   | Creation / Annihilation   | $\gamma$ with $\lambda_\gamma = 1$               | Extremal f.p.                              |
| III  | Subtle Doubling / Halving | $\lambda_\gamma = -1$                            | Transitional f.p.                          |
| IV   | Materialization / Murder  | $\lambda_\gamma = -1$                            | Transitional f.p.                          |
| V    | Phantom Kiss              | $\lambda_\gamma^3 = 1$<br>$\lambda_\gamma^4 = 1$ | 3-bifurcation point<br>4-bifurcation point |
| VI   | Emission / Absorption     | $\lambda_\gamma^k = 1$ ( $k \geq 4$ )            | $k$ -bifurcation point                     |
| VII  | Bubble / Liberation       | Perturbation of<br>critical point                | TBA                                        |
| VIII | Resonance                 | Perturbation of<br>critical point                | TBA                                        |

# I. Burst / Reincarnation

**Target.** Critical point  $x$  with pure imaginary characteristic exponents.

**Case 1.**  $\{\pm i\alpha, \pm r\}$  with  $\alpha, r \in \mathbb{R}$ . (saddle-center)



**Figure 10:** Phantom burst

# I. Burst / Reincarnation

**Target.** Critical point  $x$  with pure imaginary characteristic exponents.

**Case 1.**  $\{\pm i\alpha, \pm r\}$  with  $\alpha, r \in \mathbb{R}$ . (saddle-center)



- By Lyapunov's theorem, we have a family of orbits near  $x$
- The characteristic multipliers of the Lyapunov orbits near  $x$  is eventually real, so the orbits are hyperbolic.
- This hyperbolic family cannot be observed in the phase portrait, so we call this a **phantom burst**.

**Figure 10:** Phantom burst



# I. Burst / Reincarnation

**Case 2.**  $\{\pm i\alpha, \pm i\beta\}$  with  $\mathbb{Q}$ -independent  $\alpha, \beta \in \mathbb{R}$ . (pure center)



**Figure 11:** Stable burst



**Figure 12:** Reincarnation

# I. Burst / Reincarnation

**Case 2.**  $\{\pm i\alpha, \pm i\beta\}$  with  $\mathbb{Q}$ -independent  $\alpha, \beta \in \mathbb{R}$ . (pure center)



**Figure 11:** Stable burst

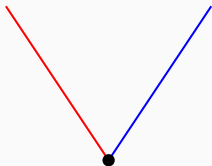


**Figure 12:** Reincarnation

- By Lyapunov's theorem, we have two families near  $\odot$ .
- The characteristic multipliers are unimodular, so both orbits are elliptic.
- If two orbit cylinders are on the same side respect to the energy hyperplane, we observe two families of orbits burst from the critical point. (**Stable burst**)
- Otherwise, we observe a family of orbit absorbed into the critical point and burst out again. (**Reincarnation**)

## II. Creation / Annihilation

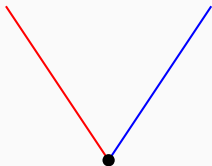
**Target.** Periodic orbit such that  $\lambda_\gamma = 1$ .



**Figure 13:** Creation and annihilation  
(Also called birth-death)

## II. Creation / Annihilation

**Target.** Periodic orbit such that  $\lambda_\gamma = 1$ .

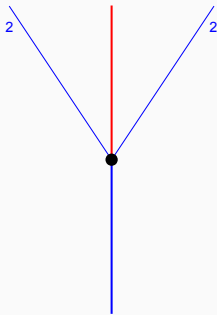


**Figure 13:** Creation and annihilation  
(Also called birth-death)

- The vector field develops an unstable periodic orbit  $\bullet$ . It's a closing of a recurrent orbit, called **Pugh catastrophe**.
- $\bullet$  splits into a hyperbolic orbit and an elliptic orbit.
- In the phase portrait, we observe an elliptic orbit suddenly appeared. (**Creation**)
- If we read this in reversed time, we can observe a vanishing pair of orbits. (**Annihilation**)

### III. Subtle Doubling / Subtle Halving

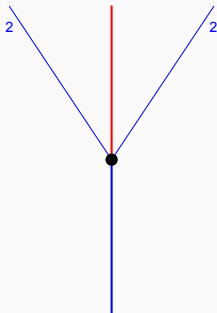
**Target.** Periodic orbit such that  $\lambda_\gamma = -1$ .



**Figure 14:** Subtle doubling and halving  
(Also called period doubling)

### III. Subtle Doubling / Subtle Halving

**Target.** Periodic orbit such that  $\lambda_\gamma = -1$ .

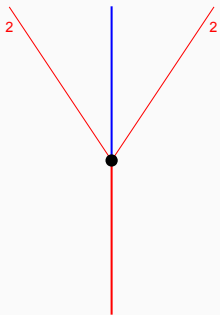


**Figure 14:** Subtle doubling and halving  
(Also called period doubling)

- The elliptic orbit cylinder becomes hyperbolic at  $\bullet$ .
- An elliptic orbit with period 2 emerges.
- In phase portrait, we observe that the period of — is doubled. (**Subtle doubling**)
- In the opposite direction, the period is halved. (**Subtle halving**)

## IV. Materialization / Murder

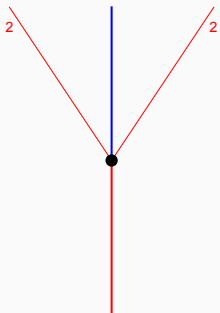
**Target.** Periodic orbit such that  $\lambda_\gamma = -1$ .



**Figure 15:** Materialization and murder

## IV. Materialization / Murder

**Target.** Periodic orbit such that  $\lambda_\gamma = -1$ .



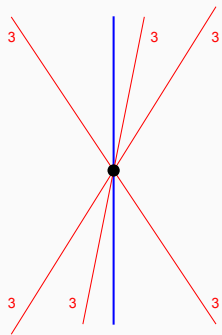
- Similar to subtle doubling, but with an interchange of elliptic and hyperbolic.
- We observe that — suddenly appeared. (**Materialization**)
- In the opposite direction, — suddenly disappears. (**Murder**)

**Figure 15:** Materialization and murder



## V. Phantom Kiss

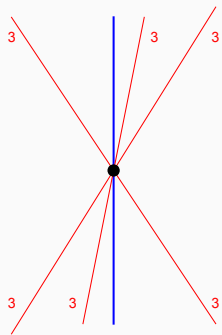
**Target.**  $\lambda_\gamma = \exp(2\pi il/k)$ ,  $(l, k) = 1$  and  $k = 3$  or  $4$ .



**Figure 16:** Phantom kiss

## V. Phantom Kiss

**Target.**  $\lambda_\gamma = \exp(2\pi il/k)$ ,  $(l, k) = 1$  and  $k = 3$  or  $4$ .

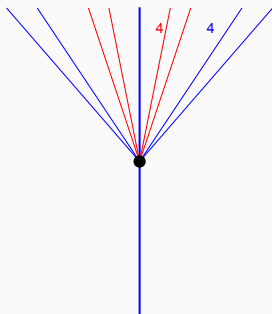


- A hyperbolic orbit of period 3 emerges in both directions of  $\bullet$ .
- We only observe *a moment of instability* of  $\text{—}$ .
- The family of hyperbolic orbit only experiences a sudden absorption to the elliptic orbit. **(Phantom kiss)**

**Figure 16:** Phantom kiss

## VI. Emission / Absorption

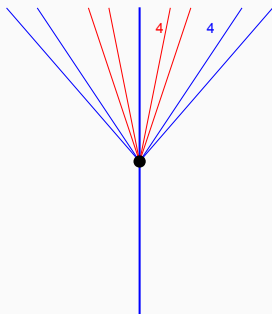
**Target.**  $\lambda_\gamma = \exp(2\pi i l/k)$ ,  $(l, k) = 1$  and  $k \geq 4$ .



**Figure 17:** Emission and absorption

## VI. Emission / Absorption

**Target.**  $\lambda_\gamma = \exp(2\pi i l/k)$ ,  $(l, k) = 1$  and  $k \geq 4$ .



- A hyperbolic and an elliptic orbit of period  $k$  both emerges at  $\bullet$ . (**Emission**)
- An elliptic orbit of period  $k$  is observed.
- In the opposite direction,  $k$ -periodic orbits are absorbed into the original periodic orbit. (**Absorption**)

**Figure 17:** Emission and absorption

## VII. Bubble / Liberation

**Target.** Fixed point 0 of parametrized Hamiltonian  $H_\mu$ .

Assume that 0 is elliptic if  $\mu < 0$  and hyperbolic if  $\mu > 0$ , i.e.

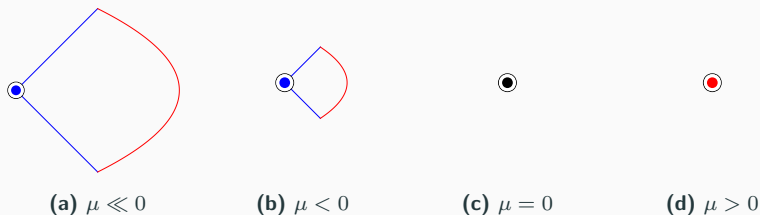
- $\mu < 0$  :  $CE(x) = \{\pm i\alpha, \pm i\beta\}$  for  $\alpha, \beta \in \mathbb{R}_{>0}$ ,  $\alpha \neq \beta$ .
- $\mu = 0$  :  $\alpha = \beta$
- $\mu > 0$  :  $CE(x) = \{\pm z, \pm \bar{z}\}$  where  $z \notin S^1$ .

If  $\mu < 0$ , the stable burst occurs, while nothing happens if  $\mu > 0$ .

**Question.** What happens to two elliptic orbit cylinders?

[MS71] showed that there are only two possibilities in general.

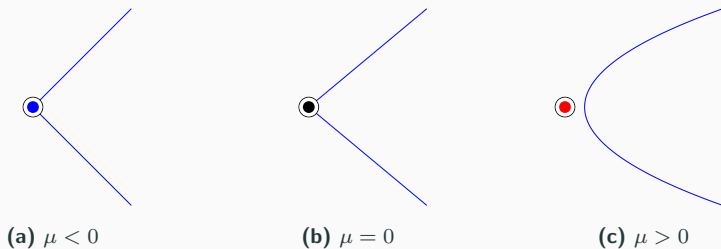
## VII. Bubble / Liberation



**Figure 18:** Bubble

The first possibility is a **bubble**: the shrink of two orbit cylinders into a fixed point.

## VII. Bubble / Liberation



**Figure 19:** Liberation

The second possibility is a **liberation**: two orbit cylinders patch together. This phenomenon actually happens in CRTBP at  $L_4$ .

## VIII. Resonance

**Target.** Fixed point 0 of parametrized Hamiltonian  $H_\mu$ .

For characteristic exponents  $\alpha_1$  and  $\alpha_2$ , if there exist coprime integers  $k, l$  such that

$$k\alpha_1 = l\alpha_2,$$

then  $\mathcal{H}$ -elementary assumption of Lyapunov theorem is violated.

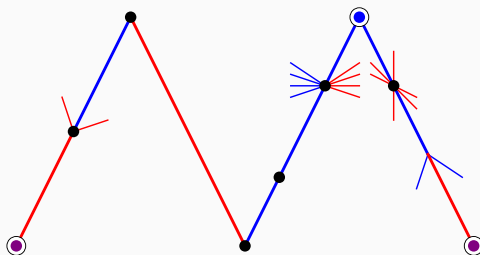
This is called **resonance**, and there are many possible different phenomenon occurs depending on  $(k, l)$ .

[SS73] gives the classification of generic resonance.

The situation is quite similar to the case of the bubble: ‘bridges’ between two orbit cylinders are collapsed into a point.



# A Life of Orbit



**Figure 20:** A life of an orbit

A periodic orbit can be born and die at critical point or other orbit.

For varying energy, it experiences various bifurcations, make birth of another orbits, and become from hyperbolic to elliptic or vice versa.

## Closing

---

# Genericity and Symmetry

Generically, a given Hamiltonian system is generic. (It's trivial.)

# Genericity and Symmetry

Generically, a given Hamiltonian system is generic. (It's trivial.)

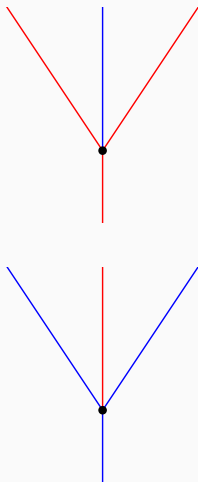
However, most of interesting systems are *not* generic, because of their symmetries in nature.

- The circular restricted three-body problem has  $\mathbb{Z}_2 \times \mathbb{Z}_2$ -symmetry.
- [Ayd23] The linear symmetries of Hill's lunar problem, a limit case of of CRTBP, is given by  $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ .
- [AFFvK13], [Lee25] There are much more linear symmetries in the rotating Kepler problem, so there is  $S^1$ -family bifurcation in the planar case, and  $S^3$ -family bifurcation in the spatial case.

Such symmetries prevent the bifurcation from being generic.

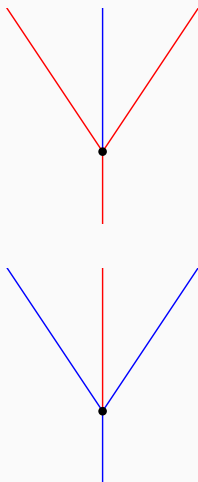
Many works were done for symmetric Hamiltonians, including [GSM87].

# Pitchfork Bifurcation



**Figure 21:** Pitchfork bifurcations

# Pitchfork Bifurcation



**Figure 21:** Pitchfork bifurcations

- The orbit changes hyperbolicity and ellipticity, and **two** orbits with **period 1** are born.
- The reason is  $\mathbb{Z}_2$ -symmetry; if there is a new orbit one side, then there must be another one on the other side.
- This type of bifurcation, which is not *generic*, is observed in Hill's lunar problem, CRTBP, and many other systems with symmetry.

## Further Questions

Investigating bifurcation by Conley-Zehnder index:

- $SH$  provides systematic way to study bifurcation.
- [AFvK<sup>+</sup>] studied Hill's lunar problem with CZ index.
- We might apply the method to three-body problem and other interesting systems to investigate periodic orbits and bifurcations.

## Further Questions

Investigating bifurcation by Conley-Zehnder index:

- $SH$  provides systematic way to study bifurcation.
- [AFvK<sup>+</sup>] studied Hill's lunar problem with CZ index.
- We might apply the method to three-body problem and other interesting systems to investigate periodic orbits and bifurcations.

General relation between the change of index and the bifurcation type:

- If we know the bifurcation type, what can we say about the index?
- If we detect index change, what can we say about bifurcation type?
- [FKM23] showed the invariance of  $\chi_{SFT}$  explicitly in each case.



# The End



**Figure 22:** Thank you for your attention!

## Appendix 1. Twist Condition

$\Psi : 0 \in U \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is  $\alpha$ -**twist map** if 0 is a fixed point of  $\Psi$  with characteristic exponents  $\pm i\alpha$  for  $\alpha \in \mathbb{R}$ .

If we can write

$$\Psi(z) = e^{i\alpha + \beta|z|}z + R_4(z),$$

where  $R_4(z) \leq K|z|^4$  for some  $K > 0$ , we call such  $\Psi$  is written in  $(\alpha, \beta)$ -**normal form**.

### Theorem (Birkhoff-Sternberg-Moser Normal Form)

*Let  $\Psi$  be  $\alpha$ -twist mapping with  $\alpha \neq 0, k\pi/2$  or  $2k\pi/3$ . Then there exists a chart such that  $\Psi$  can be written in  $(\alpha, \beta)$ -normal form and the sign of  $\beta$  is independent of the change of the chart.*

An  $\alpha$ -twist map is an **elementary twist map** if  $\alpha \neq 0, k\pi/2$ , or  $2k\pi/3$  and  $\beta \neq 0$ , which was mentioned in the Birkhoff-Lewis theorem.

## Appendix 2. Fixed point and Generating Function

Let the 2nd-order part of  $G$  be

$$G(\xi, \eta) = \frac{1}{2}(\alpha\xi^2 + 2\beta\xi\eta + \gamma\eta^2),$$

and  $4\Delta = \alpha\gamma - \beta^2 \neq -1$ . The linear part of  $(Q, P)$  is given by

$$\begin{pmatrix} Q \\ P \end{pmatrix} = \frac{1}{1 + 4\Delta} \begin{pmatrix} (1 - \beta)^2 - \alpha\gamma & -2\gamma \\ 2\alpha & (1 + \beta)^2 - \alpha\gamma \end{pmatrix} \begin{pmatrix} q \\ p \end{pmatrix}.$$

The trace of given matrix is  $T = 2(1 - 4\Delta)(1 + 4\Delta)^{-1}$ .

- Min/Max of  $G \Leftrightarrow \Delta > 0 \Leftrightarrow |T| < 2 \Leftrightarrow$  Elliptic fixed point
- Saddle of  $G \Leftrightarrow \Delta < 0 \Leftrightarrow |T| > 2 \Leftrightarrow$  Hyperbolic fixed point

## Appendix 3. Non-degeneracy of bifurcation points

A fixed point  $(p, s)$  is **transitional** if locally

$$\begin{pmatrix} Q \\ P \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} q \\ p \end{pmatrix} + (\text{h.o.t}),$$

and for

$$P_{qq} = 2\alpha, \quad P_{qp} = \beta, \quad P_{qt} = \gamma, \quad P_{qqq} = 6\delta, \quad Q_{qq} = 2\zeta$$

with  $\eta = \delta + \alpha\zeta + \alpha\beta/2$ ,

$$\gamma \neq 0, \quad \alpha^2 + 2\eta \neq 0.$$

## Appendix 3. Non-degeneracy of bifurcation points

Let  $CM(p) = \{\exp(\pm 2\pi i l/k)\}$ ,  $k \geq 3$ ,  $(l, k) = 1$  and  $0 < l < k$ .

[Bir27] introduced a normalization procedure, which gives local expression

$$\begin{aligned}(q, p) &= (r^{1/2} \cos \theta, r^{1/2} \sin \theta), \\(Q, P) &= (R^{1/2} \cos \Theta, R^{1/2} \sin \Theta), \\ \Theta &= \theta + \frac{2\pi l}{k} + \frac{t\alpha(t)}{k} + \sum_{j=1}^{\lfloor (k-2)/2 \rfloor} \frac{\beta_j(t)}{k} r^j \\ &\quad + \frac{\gamma(t)}{k} r^{(k-2)/2} \cos k\theta + \Theta_1(\theta, r, t), \\ R &= r + \frac{2\gamma(t)}{k} r^{k/2} \sin k\theta + R_1(\theta, r, t).\end{aligned}$$

## Appendix 3. Non-degeneracy of bifurcation points

Moreover, we require for  $r^{1/2} = \rho$ ,

$$\frac{\partial^j R_1}{\partial \rho^j}(\theta, 0, 0) = 0 \quad \text{for } j = 0, 1, \dots, k+1$$

$$\frac{\partial^i \Theta_1}{\partial \rho^i}(\theta, 0, 0) = 0 \quad \text{for } i = 0, 1, \dots, k-1.$$

We denote  $\alpha(0) = \alpha$ ,  $\beta_1(0) = \beta$ ,  $\gamma(0) = \gamma$ . If

- $(k = 3) \ \alpha, \gamma \neq 0$ ,
- $(k = 4) \ \alpha, \gamma, \beta \pm \gamma \neq 0$ ,
- $(k \geq 5) \ \alpha, \beta, \gamma \neq 0$ ,

we call  $(p, s)$  a  **$k$ -bifurcation point**.

## References i

-  Peter Albers, Joel W. Fish, Urs Frauenfelder, and Otto van Koert, *The Conley-Zehnder indices of the rotating Kepler problem*, Math. Proc. Cambridge Philos. Soc. **154** (2013), no. 2, 243–260. MR 3021812
-  Cengiz Aydin, Urs Frauenfelder, Otto van Koert, Dayoung Koh, and Agustin Moreno, *Symplectic geometry and space mission design*.
-  Ralph Abraham and Jerrold E. Marsden, *Foundations of mechanics*, second ed., Benjamin/Cummings Publishing Co., Inc., Advanced Book Program, Reading, MA, 1978, With the assistance of Tudor Rațiu and Richard Cushman. MR 515141
-  Cengiz Aydin, *The linear symmetries of Hill's lunar problem*, Arch. Math. (Basel) **120** (2023), no. 3, 321–330. MR 4549765
-  G.D. Birkhoff, *Dynamical systems*, American Mathematical Society / Providence, Estados Unidos, American Mathematical Society, 1927.

## References ii



G. D. Birkhoff and D. C. Lewis, *On the periodic motions near a given periodic motion of a dynamical system*, Ann. Mat. Pura Appl. **12** (1934), no. 1, 117–133. MR 1553217



Michael A. Buchner, *On the generic nature of property  $H1$  for Hamiltonian vector fields*, Global Analysis (Proc. Sympos. Pure Math., Vols. XIV, XV, XVI, Berkeley, Calif., 1968), Proc. Sympos. Pure Math., vol. XIV-XVI, Amer. Math. Soc., Providence, RI, 1970, pp. 51–54. MR 264701



M.S El Naschie, *Vak, vacuum fluctuation and the mass spectrum of high energy particle physics*, Chaos, Solitons & Fractals **17** (2003), no. 4, 797–807.



## References iii

-  Urs Frauenfelder, Dayung Koh, and Agustin Moreno, *Symplectic methods in the numerical search of orbits in real-life planetary systems*, SIAM Journal on Applied Dynamical Systems **22** (2023), no. 4, 3284–3319.
-  Martin Golubitsky, Ian Stewart, and Jerrold Marsden, *Generic bifurcation of hamiltonian systems with symmetry*, Physica D: Nonlinear Phenomena **24** (1987), no. 1, 391–405.
-  Michel Hénon and Carl Heiles, *The applicability of the third integral of motion: Some numerical experiments*, Astronom. J. **69** (1964), 73–79. MR 158746
-  Dongho Lee, *Conley-zehnder indices of spatial rotating kepler problem*, 2025.
-  K. R. Meyer, *Generic bifurcation of periodic points*, Transactions of the American Mathematical Society **149** (1970), no. 1, 95–107.

## References iv

-  K. R. Meyer and D. S. Schmidt, *Periodic orbits near  $\mathcal{L}_4$  for mass ratios near the critical mass ratio of Routh*, Celestial Mech. **4** (1971), 99–109. MR 317617
-  H. Poincaré, *Les méthodes nouvelles de la mécanique céleste. tome III*, Les Grands Classiques Gauthier-Villars. [Gauthier-Villars Great Classics], Librairie Scientifique et Technique Albert Blanchard, Paris, 1987, Invariant intégraux. Solutions périodiques du deuxième genre. Solutions doublement asymptotiques. Reprint of the 1899 original, Bibliothèque Scientifique Albert Blanchard. MR 926908
-  S. Smale, *Differentiable dynamical systems*, Bull. Amer. Math. Soc. **73** (1967), 747–817. MR 228014
-  Dieter S. Schmidt and Daniel Sweet, *A unifying theory in determining periodic families for hamiltonian systems at resonance*, Journal of Differential Equations **14** (1973), no. 3, 597–609.

## References v



Ruikang Zhang, Yue Wang, Hao Zhang, and Chen Zhang, *Transfers from distant retrograde orbits to low lunar orbits*, Celestial Mechanics and Dynamical Astronomy **132** (2020), 41.